

A Sample Book Using Quarto with Standard LaTeX Book Class

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Chapter 1

Introduction

This file is a sample book demonstrating Quarto’s capabilities with the standard LaTeX book document class. It supports PDF (via LaTeX) output. It is assumed that the reader has a basic working knowledge of stochastic calculus Karatzas and Shreve [1] and stochastic control theory Yong and Zhou [2].

$$\begin{aligned} \forall x \in \mathbb{R}, \quad \exists y \in \mathbb{N}: \quad \alpha + \beta = \gamma \neq \delta \leq \epsilon \approx \zeta \\ \sum_{i=1}^n x_i^2 + \prod_{j=1}^m y_j^2 = \int_a^b f(x) \, dx \\ \lim_{n \rightarrow \infty} \frac{1}{n} = 0, \quad \nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \\ A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \\ S = \{x \mid x > 0\}, \quad f: X \rightarrow Y, \quad x \mapsto f(x) \\ \bigcup_{i=1}^n A_i \cap \bigcap_{j=1}^m B_j, \quad A \subseteq B, \quad A \supset B \\ \mathcal{L} = \mathcal{F}(\mathbb{R}), \quad \mathfrak{g} \cong \mathfrak{h} \\ \boxed{E = mc^2} \end{aligned}$$

Theorem 1.0.1 (Residue Theorem). *Let f be analytic in a simply connected domain D . Then for any closed contour $\gamma \subset D$,*

$$\oint_{\gamma} f(z) \, dz = 2\pi i \sum_k \text{Res}(f, z_k) \tag{1.1}$$

This template supports standard Quarto cross-references. You can refer to Theorem Theorem 1.0.1 in the text.

Chapter 2

Markdown Basics

The template is built on Quarto and Pandoc, (1.1) leveraging standard LaTeX tools for PDF output.

2.1 Equations

Inline math: $E = mc^2$. Display math:

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u$$

2.2 Theorems and Proofs

Lemma 2.2.1 (Key Lemma). *If f is continuous on $[a, b]$, then f is integrable on $[a, b]$.*

Corollary 2.2.1 (Easy Corollary). *Every continuous function on a closed interval is bounded.*

Definition 2.2.1 (Model Definition). A stochastic process $\{X_t\}_{t \geq 0}$ is a collection of random variables indexed by time.

2.3 Lists

1. First item
 2. Second item
 3. Third item
- Bullet one
 - Bullet two

2.4 Tables

2.4 Tables

Table 2.1: A sample table

Variable	Description	Unit
x	Position	m
v	Velocity	m/s
a	Acceleration	m/s ²

Chapter 3

Bibliography

This template uses natbib with the plainnat-doi bibliography style for author-year citations
Karatzas and Shreve [\[1\]](#).

Acknowledgment

This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this book.

Use the `[]{\appendix options="An Appendix"}` markup if you have a single appendix. Do not use `\section{}` after `\appendix` in LaTeX.

References

- [1] Ioannis Karatzas and Steven E. Shreve. *Brownian Motion and Stochastic Calculus*. Springer New York, 1991.
- [2] Jiongmin Yong and Xunyu Zhou. *Stochastic Controls: Hamiltonian Systems and HJB Equations*. Stochastic Modelling and Applied Probability. Springer New York, 1999.

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